

Question	Answer	Marks
7(a)	work done per unit charge	B1
	(work done) moving positive charge from infinity (to the point)	B1
7(b)(i)	potential always same sign / potential is always positive so same sign of charge	B1

Question	Answer	Marks
7(b)(ii)	1 <i>from x = 12 cm to x = 25 cm:</i> speed increases and <i>from x = 27 cm to x = 31 cm:</i> speed decreases	B1
	<i>(from x = 12 cm to x = 25 cm:</i> speed increases) at decreasing rate or <i>(from x = 27 cm to x = 31 cm:</i> speed decreases) at increasing rate	B1
	<i>at x = 26 cm:</i> speed maximum	B1
	<i>at 32 cm:</i> speed still decreasing	B1
	2 $q \Delta V = \frac{1}{2}mv^2$ $3.2 \times 10^{-19} \times (2.14 - 1.43) \times 10^4 = \frac{1}{2} \times 6.6 \times 10^{-27} \times v^2$ $v^2 = 6.88 \times 10^{11}$	C1
	$v = 8.3 \times 10^5 \text{ m s}^{-1}$ (8.30)	A1

Question	Answer	Marks
8(a)(i)	all frequencies have the same gain	B1